Machine Learning Models for Predicting Diabetes Risk Using Routine Population Health Indicators

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Background & Motivation

- Diabetes is a major global health burden
- Associated with complications such as cardiovascular disease, kidney failure, and neuropathy
- Early detection reduces morbidity and mortality
- Need for data-driven clinical decision support tools

Problem Statement

Traditional diagnosis relies on laboratory tests (HbA1c, fasting glucose)

Limited accessibility in large-scale or low-resource settings

Need for non-invasive, scalable prediction methods

Goal: Predict diabetes using routine population health indicators

Target Population & Justification

- Adults at risk of diabetes
- High disease burden with preventable complications
- Enables early intervention and population health planning
- Supports scalable screening approaches

Target Population & Justification

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- Adults at risk of diabetes

Justification

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Literature Review

- The WHO reports that diabetes has risen sharply worldwide and remains a major public health problem with serious complications and high mortality. [World Health Organization, 2016]
- The WHO and ADA both emphasize prevention, healthy behavior, and timely management to reduce complications and improve outcomes. [World Health Organization, 2016; American Diabetes Association, 2024]
- Fasting glucose and HbA1c testing are clinically important, but they require laboratory access and may be difficult to use in population-wide screening programs. [American Diabetes Association, 2024]
- BRFSS-based datasets provide demographic, behavioral, and health indicators that are useful for modeling diabetes risk in large populations. [Teboul, 2021]

Literature Review Cont'd

- Reviews of diabetes research show that machine learning and data mining methods are commonly applied for screening, prediction, and knowledge extraction from health data. [Kavakiotis et al., 2017]

- Logistic regression is useful for interpretability, while Random Forest and Gradient Boosting are strong ensemble methods for improving predictive performance. [Breiman, 2001; Friedman, 2001]

- Variables such as BMI, physical inactivity, age, hypertension, cholesterol, and smoking are consistently associated with diabetes risk. [World Health Organization, 2016; Teboul, 2021]

- Gap : The gap is that previous studies often used clinical or lab-based data, while this research focuses on routine, non-invasive population health indicators and compares multiple machine learning models for practical diabetes screening.

Aim and Objectives

Aim

To develop and evaluate machine learning models for predicting diabetes risk using routine population health indicators, enabling early detection and improved clinical decision-making.

Objectives

- To utilize population health data (e.g., BRFSS indicators) for diabetes risk prediction
- To implement and compare multiple machine learning models (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and XGBoost)
- To address class imbalance using appropriate techniques (e.g., class weighting)
- To evaluate model performance using key metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC)
- To identify the most effective and clinically relevant model for diabetes prediction
- To support early intervention strategies and improve population health outcomes

Dataset & Features

- Dataset: BRFSS Diabetes Health Indicators
- ~253,000 observations
- Features include:
  - BMI, age, smoking, physical activity
  - Blood pressure, cholesterol
  - General health indicators
- Target variable:
  - 1 = diabetes (including prediabetes/gestational)
  - 0 = no diabetes

Modeling Pipeline

- Data loading and validation
- Feature selection and preprocessing
- Stratified train-test split (80/20)
- Class imbalance handling (class weighting / threshold tuning)
- Model training and evaluation

Models Used & Evaluation Metrics

Models Used

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- Gradient Boosting

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Model Comparison

```
MODEL COMPARISON
                  Model  CV Mean Score  Accuracy  Precision    Recall  \
3        Random Forest       0.457842  0.761057   0.332516  0.709718
1              XGBoost       0.444988  0.739179   0.315515  0.745650
4    Gradient Boosting       0.443757  0.721539   0.306507  0.790918
0  Logistic Regression       0.443292  0.731433   0.310656  0.760928
2        Decision Tree       0.430727  0.708688   0.293857  0.777479

         F1   ROC-AUC
3  0.452859  0.822822
1  0.443407  0.819203
4  0.441802  0.827101
0  0.441191  0.819634
2  0.426509  0.812212
```

Model Improvement

- Applied class balancing techniques(SMOTE)
- Adjusted classification threshold
- Improved minority class detection
- Evaluated using confusion matrix

Final Results

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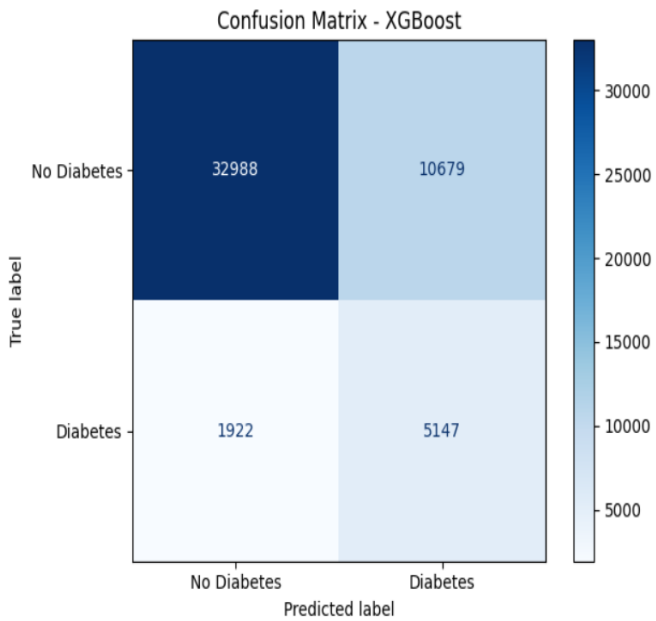
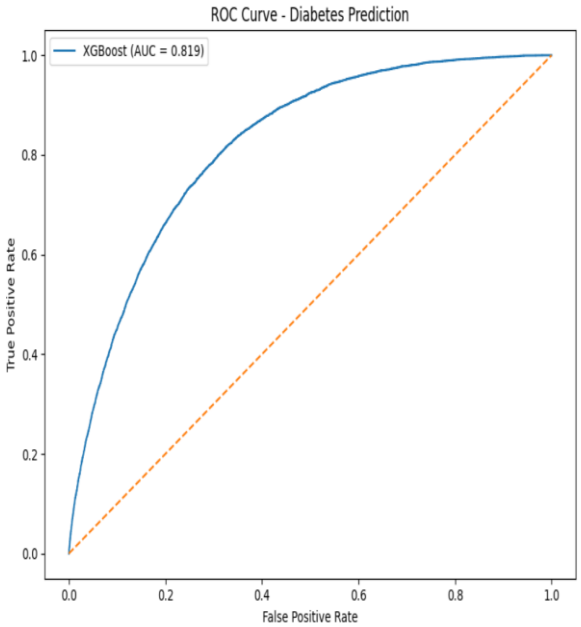
Classification Report:
              precision    recall  f1-score   support

         0.0       0.94      0.76      0.84     43667
         1.0       0.33      0.73      0.45      7069

    accuracy                           0.75     50736
   macro avg       0.64      0.74      0.64     50736
weighted avg       0.86      0.75      0.79     50736

Confusion Matrix:
[[32988 10679]
 [ 1922  5147]]
<Figure size 600x500 with 0 Axes>
```





Final Results Cont'd

- High recall supports early detection, though precision remains low
- Model effectively separates diabetic vs non-diabetic individuals
- ROC-AUC of 0.819 indicates good model discrimination ability

Clinical Interpretation

High recall (73%) means the model correctly identifies most diabetic patients. Important for screening and early intervention.

Lower precision (33%) indicates many non-diabetic individuals flagged as Diabetic. Leads to Additional testing , Increased healthcare costs , Possible patient anxiety.

Best suited as a Screening tool (not diagnostic tool). Should be followed by Laboratory confirmation (HbA1c, glucose tests)

Challenges / Limitations

- Class imbalance in dataset
- Self-reported data (recall bias, misclassification)
- Limited features (no lab-confirmed data)
- Lack of external validation

Real-World Application

- Can support clinicians in early diabetes screening
- Useful for public health surveillance programs
- Can be integrated into Clinical Decision Support Systems (CDSS)
- Enables population-level risk stratification

Future Works

IMPROVE PRECISION VIA THRESHOLD TUNNING

INCORPORATE ADDITIONAL CLINICAL DATA

DEPLOY AS WEB-BASED DECISION SUPPORT TOOL

Conclusion & Impact

- Machine learning can effectively predict diabetes risk
- Ensemble models provide strong performance
- Handling class imbalance improves clinical usefulness
- Supports early intervention and better healthcare outcomes

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Thank You